

DATASET INSPECTION REPORT

DATASET OVERVIEW

The Sample - Superstore.csv dataset contains transactional retail data covering sales orders, customers, products, geographic information, shipping details, discounts, quantities, and profitability.

The dataset contains:

- 9,994 records
- 21 columns
- 5,009 unique orders
- 793 unique customers
- 1,862 unique products
- Order dates from 3 January 2014 to 30 December 2017

DATASET STRUCTURE

Field	Data Type	Description
Row ID	Integer	Unique transaction row identifier
Order ID	Text	Order identifier
Order Date	Date*	Date the order was placed
Ship Date	Date*	Date the order was shipped
Ship Mode	Text	Shipping method
Customer ID	Text	Customer identifier
Customer Name	Text	Customer name
Segment	Text	Customer segment
Country	Text	Country
City	Text	City
State	Text	State
Postal Code	Integer	Postal code

Region	Text	Sales region
Product ID	Text	Product identifier
Category	Text	Product category
Sub-Category	Text	Product sub-category
Product Name	Text	Product description/name
Sales	Decimal	Sales revenue
Quantity	Integer	Quantity sold
Discount	Decimal	Discount applied
Profit	Decimal	Profit generated

DATA QUALITY ASSESSMENT

1. Missing Values

The inspection identified: 0 missing values

2. Duplicate Records

The dataset contained: 0 completely duplicated records

3. Data Types

Field	Initial Type	Required Type
Order Date	Text	Date
Ship Date	Text	Date

The date fields were converted to Date format using Power Query to support time-based analysis.

Numeric fields were retained as appropriate:

- Sales - Decimal
- Profit - Decimal
- Discount - Decimal
- Quantity 0 - Whole Number
- Postal Code - Whole Number

NUMERIC SUMMARY

Measure	Result
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Orders	5,009
Average Sales per Order	\$458.61
Profit Margin	12.47%
Total Transaction Rows	9,994

CATEGORY SUMMARY

Field	Unique Values
Regions	4
Customer Segments	3
Categories	3
Sub-Categories	17
States	49
Cities	531
Customers	793
Products	1,862

DATA MODELLING PREPARATION

Following data preparation, the dataset was structured for Power BI analysis using a central Fact_Sales table supported by analytical dimensions including:

- Dim_Date
- Dim_Customer
- Dim_Product

The model supports analysis across:

Time - Customer - Product - Sales/Profit

The geographic fields remain available for regional and map-based analysis.

CONCLUSION

The inspection confirms that the Superstore dataset is suitable for the required business analysis and Power BI dashboard.

No missing values or complete duplicate records were identified. The primary data preparation requirement identified was the conversion of the Order Date and Ship Date fields from text to date format.

The dataset contains sufficient information to answer the assignment's required business questions covering overall sales performance, regional performance, customer segments, product categories, product profitability, and trends over time.

The prepared dataset is ready for data modelling, DAX measure development, dashboard creation, and business insight generation.